5.4 The Rationals Inside the Reals

Section 5.3 made $\mathbb{R}$ an ordered field. We built it to enlarge $\mathbb{Q}$, but $\mathbb{Q}$ is not yet inside it: a rational is a class of pairs of integers, a real number a class of sequences. This section places a copy of $\mathbb{Q}$ in $\mathbb{R}$, preserved by the operations and the order; proves that $\mathbb{R}$ is Archimedean, so that Chapter 4's theorems about Archimedean fields apply to it, density first; and meets the number that Chapter 4 promised, a positive real whose square is 2.

Notation is as in Section 5.3: $\mathcal{C}$ is the set of Cauchy sequences of rationals, $[a]$ is the class of $a \in \mathcal{C}$, $(q)_n$ is the constant sequence with value q, $P = \mathbb{R}_{>0}$ is the set of positive reals, and $x < y$ means $y - x \in P$ (Definition 5.3.2). By Theorem 5.3.3, the results of Chapters 2 and 4 for an arbitrary ordered field apply with $F = \mathbb{R}$; we say "in $\mathbb{R}$" where the context does not make the field clear.

5.4.1 A copy of the rationals

Section 5.2 already represented each rational q by the class of its constant sequence, as Chapter 4 said it would. We make that a map.

Definition 5.4.1. The Rationals in the Reals

For $q \in \mathbb{Q}$, let

$$\kappa(q) \coloneqq [(q)_n] \in \mathbb{R},$$

the class of the constant sequence with value q. This defines a function $\kappa\colon \mathbb{Q} \to \mathbb{R}$.

The right side is in $\mathbb{R}$ because constant sequences are Cauchy (Theorem 4.5.3 (a)), and κ is a function because each q determines one constant sequence and that sequence one class. In this notation $0_{\mathbb{R}} = \kappa(0)$ and $1_{\mathbb{R}} = \kappa(1)$ (Definition 5.2.2). The letter κ is new on purpose: Chapter 2 used ι and j for $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q}$, and Chapter 4 wrote ι_F for the copy of $\mathbb{Q}$ in an ordered field F. Part (d) of the next theorem shows that κ is $\iota_{\mathbb{R}}$.

The theorem says that κ carries everything we know how to do with rationals into $\mathbb{R}$ without change: sums, products, negatives, reciprocals, the order and the absolute value.

Theorem 5.4.1. The Rationals Embed in the Reals

For all $q, s \in \mathbb{Q}$:

(a) $\kappa(q+s) = \kappa(q) + \kappa(s)$, $\kappa(qs) = \kappa(q)\kappa(s)$, $\kappa(0) = 0_{\mathbb{R}}$ and $\kappa(1) = 1_{\mathbb{R}}$;

(b) $q < s$ if and only if $\kappa(q) < \kappa(s)$; hence κ is injective;

(c) $\kappa(-q) = -\kappa(q)$, $\kappa(|q|) = |\kappa(q)|$, and, if $q \neq 0$, $\kappa(q) \neq 0_{\mathbb{R}}$ and $\kappa(1/q) = \kappa(q)^{-1}$;

(d) κ is the map $\iota_{\mathbb{R}}$ of Theorem 4.2.2 for the ordered field $F = \mathbb{R}$; in particular $\kappa(n) = n \cdot 1_{\mathbb{R}}$ for every $n \in \mathbb{N}$ (Definition 4.2.1).

Idea. Constant sequences add, multiply and negate termwise into constant sequences, so (a) and the first identity of (c) are the definitions of the operations. For (b), the difference of two constant sequences is constant, and a constant $s - q > 0$ is its own margin. The rest is algebra in a field: the absolute value by cases, reciprocals by uniqueness of inverses, and (d) by the uniqueness clause of Chapter 4's embedding theorem.

Proof.

(a) By Definition 5.2.2, $\kappa(q)+\kappa(s) = [(q)_n+(s)_n] = [(q+s)_n] = \kappa(q+s)$, since the termwise sum of two constant sequences is the constant sequence of the sum. The product is the same with $\cdot$ in place of $+$. The last two identities are the definitions of $0_{\mathbb{R}}$ and $1_{\mathbb{R}}$.
(c), first identity. By Definition 5.2.2, $-\kappa(q) = [-(q)_n] = [(-q)_n] = \kappa(-q)$.

(b) ($\Rightarrow$) Let $q < s$. By (a) and the first identity of (c), $\kappa(s) - \kappa(q) = \kappa(s) + \kappa(-q) = \kappa(s - q) = [(s - q)_n]$. The constant $s - q$ lies in $\mathbb{Q}_{>0}$ (Proposition 2.4.3 (a)), so the constant sequence $(s - q)_n$ is positive, with margin $\delta = s - q$ and threshold $N = 0$ (Definition 5.3.1). Hence $\kappa(s) - \kappa(q) \in P$, that is, $\kappa(q) < \kappa(s)$.

($\Leftarrow$) Let $\kappa(q) < \kappa(s)$. By trichotomy in $\mathbb{Q}$, $q < s$, $q = s$ or $s < q$. If $q = s$, then $\kappa(q) = \kappa(s)$; if $s < q$, then $\kappa(s) < \kappa(q)$ by ($\Rightarrow$). Either contradicts $\kappa(q) < \kappa(s)$ by (O1) in $\mathbb{R}$ (Theorem 5.3.3). So $q < s$.

Injectivity: if $q \neq s$, then $q < s$ or $s < q$, so $\kappa(q) < \kappa(s)$ or $\kappa(s) < \kappa(q)$, and in either case $\kappa(q) \neq \kappa(s)$ by (O1).

(c), the rest. If $q > 0$, then $\kappa(q) > \kappa(0) = 0_{\mathbb{R}}$ by (b), so $|\kappa(q)| = \kappa(q) = \kappa(|q|)$ (Definition 4.1.1, in $\mathbb{R}$ and in $\mathbb{Q}$). If $q = 0$, both sides are $0_{\mathbb{R}}$. If $q < 0$, then $\kappa(q) < 0_{\mathbb{R}}$ by (b), so $|\kappa(q)| = -\kappa(q) = \kappa(-q) = \kappa(|q|)$. Now let $q \neq 0$. Then $\kappa(q) \neq \kappa(0) = 0_{\mathbb{R}}$ by injectivity, and by (a),

$$\kappa(q)\,\kappa(1/q) = \kappa(q \cdot (1/q)) = \kappa(1) = 1_{\mathbb{R}}.$$

In the field $\mathbb{R}$ (Theorem 5.2.3), the inverse of $\kappa(q)$ is the only element with this property (Lemma 4.1.1 (a)), so $\kappa(1/q) = \kappa(q)^{-1}$.

(d) By Theorem 5.3.3, $\mathbb{R}$ is an ordered field, so Theorem 4.2.2 gives **exactly one** map $\iota_{\mathbb{R}} \colon \mathbb{Q} \to \mathbb{R}$ with $\iota_{\mathbb{R}}(1) = 1_{\mathbb{R}}$ that preserves sums and products. By (a), κ is such a map, so $\kappa = \iota_{\mathbb{R}}$. The same theorem states that $\iota_{\mathbb{R}}(n) = n \cdot 1_{\mathbb{R}}$ for every $n \in \mathbb{N}$.

■

By (d), part (b) and the injectivity of κ also follow from Theorem 4.2.2 with $F = \mathbb{R}$; the direct proof of (b) shows a constant sequence as its own margin.

Two more identities follow from (a) and (c):

$$\kappa(q - s) = \kappa(q) + \kappa(-s) = \kappa(q) - \kappa(s),$$
$$\kappa(q/s) = \kappa(q)\,\kappa(1/s) = \kappa(q)/\kappa(s) \quad \text{for } s \neq 0.$$

So any computation with rationals, done in $\mathbb{Q}$ and then moved into $\mathbb{R}$ by κ, gives the same answer as the computation done in $\mathbb{R}$ on the images. Part (b) adds that comparisons agree too.

5.4.2 Archimedes in the reals

Chapter 4 showed that an ordered field need not be Archimedean: in $\mathbb{Q}(t)$ the element t is larger than every $n \cdot 1$ (Example (A Non-Archimedean Ordered Field)). So the Archimedean property of $\mathbb{R}$ has to be proved. It comes from $\mathbb{Q}$, through the fact that a Cauchy sequence is bounded.

Theorem 5.4.2. The Reals Are Archimedean

$\mathbb{R}$ is an Archimedean ordered field (Definition 4.2.2): for every $x \in \mathbb{R}$ there is $n \in \mathbb{N}$ with

$x < n \cdot 1_{\mathbb{R}}$.

Idea. A representative of x is a Cauchy sequence of rationals, so all its terms lie below one rational bound M. Archimedes in $\mathbb{Q}$ puts a natural number above M; one more unit gives a fixed margin above every term, and a fixed margin is what positivity asks for (Example (Method: name the margin before the index)).

Proof. Let $x \in \mathbb{R}$, and write $x = [a]$ with $a \in \mathcal{C}$. By Proposition 4.5.2 with $F = \mathbb{Q}$, there is $M \in \mathbb{Q}_{>0}$ with $|a_k| \leq M$ for every $k \in \mathbb{N}$. By Theorem 3.2.3, there is $n \in \mathbb{N}$ with $M < n$. For every k,

$$(n+1) - a_k \geq (n+1) - M = (n - M) + 1 > 1,$$

since $a_k \leq |a_k| \leq M$ (Proposition 4.1.3 (c)), so $-M \leq -a_k$ and then $(n+1)-a_k \geq (n+1)-M$ (Proposition 2.4.3 (a), (b)), and $n - M > 0$. So the sequence $((n+1)-a_k)_k$ is positive, with margin 1 and threshold 0. It represents $\kappa(n+1)-x$, by Definition 5.2.2. Hence $x < \kappa(n+1)$, and $\kappa(n+1) = (n+1) \cdot 1_{\mathbb{R}}$ by Theorem 5.4.1 (d). Since $n+1 \in \mathbb{N}$, this proves the theorem. ■

Every result of Chapter 4 about Archimedean ordered fields now applies with $F = \mathbb{R}$. Two of them are used at once.

The integer part in $\mathbb{R}$. By Proposition 4.2.4 with $F = \mathbb{R}$, and $\iota_{\mathbb{R}} = \kappa$ (Theorem 5.4.1 (d)), for every $x \in \mathbb{R}$ there is **exactly one** $m \in \mathbb{Z}$ with

$$\kappa(m) \leq x < \kappa(m+1).$$

We call m the **integer part** of x. For a rational q, the integer part of $\kappa(q)$ is the integer part $\lfloor q \rfloor$ of Proposition 3.2.5, because κ preserves and reflects $\leq$ (Theorem 5.4.1 (b)). The word "exactly" matters: an integer part is given by a rule, so a sequence of integer parts can be defined without choosing anything (Exercise B2).

5.4.3 Between two reals there is a rational

Chapter 4 proved that the rationals are dense in every Archimedean ordered field (Theorem 4.2.5), and $\mathbb{R}$ is one.

Theorem 5.4.3. The Rationals Are Dense in the Reals

Let $x, y \in \mathbb{R}$ with $x < y$. Then there is $q \in \mathbb{Q}$ with $x < \kappa(q) < y$.

Proof. By Theorem 5.4.2, $\mathbb{R}$ is an Archimedean ordered field, so Theorem 4.2.5 with $F = \mathbb{R}$ gives $q \in \mathbb{Q}$ with $x < \iota_{\mathbb{R}}(q) < y$; and $\iota_{\mathbb{R}} = \kappa$ by Theorem 5.4.1 (d). ■

Density can also be seen directly from the representatives, with no appeal to Archimedes: write $x = [a]$, $y = [b]$, go far out, and take the midpoint of two late terms. Exercise C3 carries this

out; it shows that in this construction the rationals are dense because each real number is made of rationals.

Identification. From here on we write q for $\kappa(q)$ when $q \in \mathbb{Q}$, and $\mathbb{Q} \subseteq \mathbb{R}$. This is a convention about notation, of the kind Chapter 2 adopted twice, writing n for $\iota(n)$ and m for $j(m)$ (Proposition 2.2.3, Proposition 3.1.4). It is safe because of Theorem 5.4.1: κ is injective and preserves sums, products, negatives, reciprocals, the order and the absolute value, so a statement about rationals means the same whether it is read in $\mathbb{Q}$ or, through κ, in $\mathbb{R}$. It is not a set inclusion: the rational q is a class of pairs of integers, and $\kappa(q)$ is a class of sequences of rationals, a different set. In this notation $\kappa(1/n)$ is written $1/n$, and Proposition 4.2.3 in $\mathbb{R}$ says that for every $\varepsilon \in \mathbb{R}_{>0}$ some $1/n$, $n \in \mathbb{Z}_{>0}$, lies below ε.

5.4.4 The square root of two

Chapter 4 ended with the Newton sequence (x_n) of Example (A Rational Sequence Aimed at the Gap), Cauchy in $\mathbb{Q}$ and with no limit in $\mathbb{Q}$ (Theorem 4.6.1). Its class is the number it was aiming at.

Example. The Class of the Newton Sequence

Let (x_n) be the Newton sequence of Example (A Rational Sequence Aimed at the Gap), and let $r := [(x_n)] \in \mathbb{R}$. Show that $r > 0$, that $r^2 = 2$, that $r \neq q$ for every $q \in \mathbb{Q}$, and that $r = [(d_k)]$, where (d_k) is the sequence of decimal truncations of Example (The Truncations and the Newton Sequence Are Equivalent).

Solution. The sequence (x_n) is in $\mathcal{C}$ by Theorem 4.6.1, so $r \in \mathbb{R}$. We use the facts $x_n > 1$ for every $n \in \mathbb{N}$ and $e_n \leq 1/(4n)$ for every $n \geq 1$, where $e_n := x_n^2 - 2$, stated in Theorem 4.6.1, and $e_n > 0$ for every $n \in \mathbb{N}$, proved in Example (A Rational Sequence Aimed at the Gap). The first values are

$$e_0 = 2, \quad e_1 = \frac{1}{4}, \quad e_2 = \frac{1}{144}, \quad e_3 = \frac{1}{166464}, \quad e_4 = \frac{1}{221682772224}.$$

$r > 0$. Since $x_n > 1$ for every n, the sequence (x_n) is positive, with margin 1 and threshold 0. So $r \in P$, that is, $r > 0$. By Lemma 5.3.4, also $r \geq 1$.

$r^2 = 2$. By Definition 5.2.2, $r^2 = r \cdot r = [(x_n^2)]$, and $2 = [(2)_n]$. Termwise, $(x_n^2) - (2)_n = (e_n)$. For $n \geq 1$,

$$0 < e_n \leq \frac{1}{4n} \leq \frac{1}{n+1},$$

where the last step is Proposition 4.1.2 (c), since $4n - (n+1) = 3n - 1 > 0$. The sequence $(1/(n+1))$ is defined for every $n \in \mathbb{N}$ and null in $\mathbb{Q}$ (Example (First Limits) (b)); so is $(0)_n$. The squeeze Proposition 4.4.4 (c) in $\mathbb{Q}$, with $N_0 = 1$, shows that (e_n) is null. (At $n = 0$ the bound $1/(4n)$ is undefined and $e_0 = 2$; the threshold $N_0 = 1$ skips it.) Hence $(x_n^2) \sim (2)_n$ and $r^2 = 2$.

r is not rational. Suppose $r = \kappa(q)$ for some $q \in \mathbb{Q}$. Then $\kappa(q^2) = \kappa(q)^2 = r^2 = \kappa(2)$ by Theorem 5.4.1 (a), so $q^2 = 2$ by injectivity. This contradicts Theorem 3.3.1. So $r \notin \kappa(\mathbb{Q})$.

$r = [(d_k)]$. By Example (The Truncations and the Newton Sequence Are Equivalent), $(d_k) \sim (x_k)$, so the classes are equal.

So $\mathbb{R}$ contains a positive element whose square is 2, and $\mathbb{Q}$ contains none. It is the number Chapter 4 promised, and the decimal truncations name the same number as the Newton sequence.

Warning. r is **not** "the limit of (x_n) in $\mathbb{Q}$": there is no such limit (Theorem 4.6.1), and nothing in this section changes that. r is a new element, a class of sequences. Whether the x_n converge to it is a question about $\mathbb{R}$, with tolerances in $\mathbb{R}_{>0}$; Exercise B3 answers it.

Quick check. Find a rational number strictly between $[(d_k)]$ and $[(d_k + 1/(k+1))]$.

Solution. There is none, and Theorem 5.4.3 does not apply. The two sequences differ by $(1/(k+1))$, which is null, so they are equivalent and the classes are equal. The hypothesis $x < y$ fails, and no number lies strictly between a number and itself.

5.4.5 Exercises

5.4.5.1 A. Check your understanding

Exercise. A1

(a) Show that $\kappa(q^k) = \kappa(q)^k$ for every $q \in \mathbb{Q}$ and $k \in \mathbb{N}$, the powers being those of Definition 4.2.1 in $\mathbb{Q}$ and in $\mathbb{R}$.

(b) Let r be as in Example (The Class of the Newton Sequence). Find the integer parts of r and of $10r$, and justify each from the terms d_k: use $d_1 = 7/5$, $d_2 = 141/100$, and the bounds (T1) and (T2) stated before Example (The Truncations and the Newton Sequence Are Equivalent).

(c) True or false: if $a \in \mathcal{C}$, $q \in \mathbb{Q}$ and $a_n < q$ for every n, then $[a] < q$. Give a reason or a counterexample.

(d) Is $\kappa\colon \mathbb{Q} \to \mathbb{R}$ surjective?

Solution.

(a) By induction on k (Logic and Proof, Axiom 4.2.1, read on $\mathbb{N}$ through Corollary 1.4.3 (b)), applied to the property "$\kappa(q^k) = \kappa(q)^k$". Base: $\kappa(q^0) = \kappa(1) = 1_{\mathbb{R}} = \kappa(q)^0$, by Definition 4.2.1 (b) in $\mathbb{Q}$ and in $\mathbb{R}$ and Theorem 5.4.1 (a). Step: if $\kappa(q^k) = \kappa(q)^k$, then

$$\kappa(q^{k+1}) = \kappa(q^k q) = \kappa(q^k)\,\kappa(q) = \kappa(q)^k\,\kappa(q) = \kappa(q)^{k+1},$$

by the defining equation of powers, Theorem 5.4.1 (a), and the induction hypothesis. In particular $\kappa(q^2) = \kappa(q)^2$, the step used in Example (The Class of the Newton Sequence).

(b) By (T1) and (T2), stated before Example (The Truncations and the Newton Sequence Are Equivalent), (d_k) is non-decreasing, and $d_j < d_k + 1/10^k$ whenever $j \geq k$. With

$k = 2$: for every $j \geq 2$,

$$\frac{7}{5} = d_1 \leq d_j < d_2 + \frac{1}{100} = \frac{142}{100} = \frac{71}{50}.$$

By Lemma 5.3.4 with $N_0 = 2$, and $r = [(d_k)]$ (Example (The Class of the Newton Sequence)), $7/5 \leq r \leq 71/50$. Since $1 \leq 7/5$ and $71/50 < 2$, $1 \leq r < 2$, so the integer part of r is 1. Multiplying by $10 > 0$ in $\mathbb{R}$ (Proposition 2.4.3 (c), with equality allowed), $14 \leq 10r \leq 71/5 < 15$, so the integer part of $10r$ is 14.

(c) False. Take $a_n := q - 1/(n+1)$. Here $a \in \mathcal{C}$: $(q)_n$ is constant and $(1/(n+1))$ is null (Example (First Limits) (b)), so both are Cauchy (Theorem 4.5.3 (a)), and so is their difference (Theorem 4.5.3 (b)). Then $a_n < q$ for every n, and $a - (q)_n = (-1/(n+1))$ is null (Theorem 4.4.3 (b)), so $[a] = \kappa(q)$, and $[a] < \kappa(q)$ fails. The correct conclusion is $[a] \leq q$ (Lemma 5.3.4); see the warning after it.

(d) No. By Example (The Class of the Newton Sequence), $r \in \mathbb{R}$ is not $\kappa(q)$ for any $q \in \mathbb{Q}$.

5.4.5.2 B. Practice

Exercise. B1, used later: Rational tolerances suffice

Prove:

(a) For every $\varepsilon \in \mathbb{R}_{>0}$ there is $N \in \mathbb{Z}_{>0}$ with $\kappa(1/n) < \varepsilon$ for every $n \in \mathbb{N}$ with $n \geq N$.

(b) Let $(x_k)_{k\in\mathbb{N}}$ be a sequence in $\mathbb{R}$ and $x \in \mathbb{R}$. Then $x_k \to x$ in $\mathbb{R}$ (Definition 4.3.2 with $F = \mathbb{R}$) if and only if, **for every** $n \in \mathbb{Z}_{>0}$, **there is** $K \in \mathbb{N}$ with $|x_k - x| < \kappa(1/n)$ **for every** $k \geq K$. Likewise, (x_k) is Cauchy in $\mathbb{R}$ (Definition 4.5.1 with $F = \mathbb{R}$) if and only if, for every $n \in \mathbb{Z}_{>0}$, there is $K \in \mathbb{N}$ with $|x_j - x_k| < \kappa(1/n)$ for all $j, k \geq K$.

Solution.

(a) Let $\varepsilon \in \mathbb{R}_{>0}$. Since $\mathbb{R}$ is Archimedean (Theorem 5.4.2), Proposition 4.2.3 in $\mathbb{R}$ gives $N \in \mathbb{Z}_{>0}$ with $1/(n \cdot 1_{\mathbb{R}}) < \varepsilon$ for every $n \geq N$. For such n, $n \neq 0$, $n \cdot 1_{\mathbb{R}} = \kappa(n)$ (Theorem 5.4.1 (d)) and $1/\kappa(n) = \kappa(1/n)$ (Theorem 5.4.1 (c)). So $\kappa(1/n) < \varepsilon$.

(b) ($\Rightarrow$) For $n \in \mathbb{Z}_{>0}$, $1/n > 0$ in $\mathbb{Q}$ (Proposition 3.2.2 (a)), so $\kappa(1/n) > \kappa(0) = 0_{\mathbb{R}}$ (Theorem 5.4.1 (b)): it is one of the tolerances of the definition, which then supplies K.

($\Leftarrow$) Let $\varepsilon \in \mathbb{R}_{>0}$. By (a), there is $N \in \mathbb{Z}_{>0}$ with $\kappa(1/N) < \varepsilon$. The hypothesis with $n := N$ gives K with $|x_k - x| < \kappa(1/N)$ for every $k \geq K$, and then $|x_k - x| < \varepsilon$ by (O2). So K serves for ε. The Cauchy version is the same argument with $|x_j - x_k|$ for $j, k \geq K$ in place of $|x_k - x|$.

In practice: to prove a limit in $\mathbb{R}$, fix $n \in \mathbb{Z}_{>0}$ and find a threshold for the rational tolerance $1/n$. The threshold may depend on n, and on nothing chosen later.

Exercise. B2, used later

Let $x \in \mathbb{R}$. For $k \in \mathbb{N}$, let m_k be the integer part of $(k+1)x$, and $q_k := m_k/(k+1) \in \mathbb{Q}$. Prove that $q_k \leq x < q_k + 1/(k+1)$ for every k, and that $q_k \to x$ in $\mathbb{R}$.

Solution. Fix $k \in \mathbb{N}$. By the definition of the integer part (after Theorem 5.4.2), $m_k \leq (k+1)x < m_k+1$. Since $k+1 > 0$, also $1/(k+1) > 0$ (Proposition 3.2.2 (a)), and multiplying by it in $\mathbb{R}$ preserves both inequalities (Proposition 2.4.3 (c), with equality allowed on the left):

$$q_k = \frac{m_k}{k+1} \leq x < \frac{m_k+1}{k+1} = q_k + \frac{1}{k+1}.$$

So $0 \leq x - q_k < 1/(k+1)$ for every k. In $\mathbb{R}$, the sequence $(1/(k+1))$, that is $(1/((k+1)\cdot 1_{\mathbb{R}}))$ by Theorem 5.4.1 (c), (d), is null, by Proposition 4.3.1 and Theorem 5.4.2. The squeeze Proposition 4.4.4 (c) in $\mathbb{R}$, with lower sequence (0), gives that $(x - q_k)$ is null. Then $(q_k - x)$ is null (Theorem 4.4.3 (b)), so $q_k \to x$ (Proposition 4.3.3).

Each q_k is given by a rule: the integer part of $(k+1)x$ is the **only** integer with the defining property, so nothing is chosen.

Exercise. B3, used later: Put the proof in order

Let $a \in \mathcal{C}$. We prove that $\kappa(a_k) \to [a]$ in $\mathbb{R}$: **a Cauchy sequence of rationals converges in $\mathbb{R}$ to its own class.** The lines below, written with the identification (q for $\kappa(q)$), contain the proof, shuffled, together with two lines that do not belong. Put the correct lines in order, and say what is wrong with each of the other two.

(i) so $|[a] - a_k| = [(|a_m - a_k|)_m]$, by Exercise (C1, used later) (b);
(ii) since $a \in \mathcal{C}$, fix N with $|a_m - a_k| < 1/(2n)$ for all $m, k \geq N$;
(iii) its terms are $< 1/(2n)$ for $m \geq N$, so $|[a] - a_k| < 1/(2n)$;
(iv) so $|[a] - a_k| \leq 1/(2n) < 1/n$, and $K = N$ serves;
(v) let $k \geq N$;
(vi) let $n \in \mathbb{Z}_{>0}$; by Exercise (B1, used later: Rational tolerances suffice) (b), it suffices to find K with $|[a]-a_k| < 1/n$ for every $k \geq K$, since $|a_k-[a]| = |[a]-a_k|$ (Proposition 4.1.3 (b));
(vii) $[a] - a_k$ is represented by $(a_m - a_k)_m$;
(viii) for each k, fix N_k with $|a_m - a_k| < 1/(2n)$ for every $m \geq N_k$;
(ix) its terms are $\leq 1/(2n)$ for $m \geq N$, so $|[a] - a_k| \leq 1/(2n)$, by Lemma 5.3.4.

Solution. *Order:* (vi), (ii), (v), (vii), (i), (ix), (iv). No other order is admissible: line (ii) introduces N, which line (v) needs; line (v) fixes the k that lines (vii), (i) and (ix) are about; line (i) rewrites the class that line (vii) names; line (ix) bounds that class; and line (iv) concludes from line (ix), so it comes last.

Line (iii) uses the strict version of Lemma 5.3.4, which is false: termwise $<$ passes to classes only as $\leq$ (Example (Termwise Order Is Not an Order on Classes)). Concretely, for $a_m = 1/2 - 1/(m+2)$, $n = 1$ and $k = 0$, every term $|a_m - a_0|$ is $< 1/2$, yet $|[a] - a_0| = 1/2$. That is why the proof aims at $1/(2n)$ and then uses $1/(2n) < 1/n$.

Line (viii) claims a threshold N_k for every k, and the Cauchy property does not supply one for an early k: a_k can be far from all later terms. For $a = (5, 0, 0, \dots) \in \mathcal{C}$, $n = 1$ and $k = 0$, $|a_m - a_0| = 5$ for every $m \geq 1$, so no N_0 exists. The Cauchy property gives $|a_m - a_k| < 1/(2n)$ only when both m and k are at least one N, which is why line (v) takes $k \geq N$. (If line (viii) did hold for every k, the remaining lines would give the bound for every k, so it is the claim

that is unjustified, not the dependence on k.)

Why $1/(2n)$ and not $1/n$ directly? The Cauchy property gives $|a_m - a_k| < 1/n$, but Lemma 5.3.4 passes only the non-strict inequality to classes, so we would reach $|[a]-\kappa(a_k)| \leq \kappa(1/n)$, one symbol short of the definition. Halving the tolerance first leaves room for the strict inequality.

So every real number is a limit of rationals, namely of the terms of any of its representatives. Equivalent sequences have the same class, so their terms converge in $\mathbb{R}$ to the same number; by uniqueness of limits (Theorem 4.3.4 in $\mathbb{R}$), that number is the only limit they have. A real number is what all its representatives converge to.

5.4.5.3 C. Going deeper

Exercise. C1

(a) Let $x \in \mathbb{R}$ with $|x| \leq 1/n$ for every $n \in \mathbb{Z}_{>0}$. Prove that $x = 0$.

(b) Where does your proof use that $\mathbb{R}$ is Archimedean? Show that the statement of (a) fails in the ordered field $\mathbb{Q}(t)$ of Example (A Non-Archimedean Ordered Field).

Solution.

(a) Suppose that $|x| > 0$. Since $\mathbb{R}$ is Archimedean (Theorem 5.4.2), Proposition 4.2.3 in $\mathbb{R}$ with $\varepsilon := |x|$ gives $N \in \mathbb{Z}_{>0}$ with $1/N < |x|$. The hypothesis with $n = N$ gives $|x| \leq 1/N$. These contradict (O1). So $|x| > 0$ is false. Since $|x| \geq 0$ (Proposition 4.1.3 (a)), trichotomy leaves $|x| = 0$, and then $x = 0$ by Proposition 4.1.3 (a).

(b) Proposition 4.2.3 applies to $\mathbb{R}$ because of Theorem 5.4.2. In $\mathbb{Q}(t)$, Example (A Non-Archimedean Ordered Field) shows $0 < 1/t < 1/(n \cdot 1)$ for every $n \geq 1$. So $x := 1/t$ satisfies $|x| = x \leq 1/(n \cdot 1)$ for every $n \geq 1$, and $x \neq 0$. The statement of (a) is a form of the Archimedean property, and it fails without it.

Exercise. C2

Let r be as in Example (The Class of the Newton Sequence). Prove that the set $\{x \in \mathbb{R} : x \notin \kappa(\mathbb{Q})\}$ of irrational real numbers is dense in $\mathbb{R}$: for all $x, y \in \mathbb{R}$ with $x < y$ there is an irrational z with $x < z < y$.

Solution (Hint 1). Use Theorem 5.4.3, and the irrational number r.

Solution (Hint 2). Let $q \in \mathbb{Q}$ with $x < q < y$ (Theorem 5.4.3). How small must r/k be for $q + r/k$ to stay below y?

Solution. Let $x < y$. By Theorem 5.4.3, fix $q \in \mathbb{Q}$ with $x < q < y$. Then $y - q > 0$, and $r > 0$ (Example (The Class of the Newton Sequence)), so $\varepsilon := (y - q)/r \in \mathbb{R}_{>0}$ (Proposition 3.2.2

(a), (O4)). By Proposition 4.2.3 in $\mathbb{R}$ (Archimedean by Theorem 5.4.2), there is $k \in \mathbb{Z}_{>0}$ with $1/k < \varepsilon$. Multiplying by $r > 0$ (Proposition 2.4.3 (c)), $r/k < y - q$. Let $z := q + r/k$. Since $r/k > 0$,

$$x < q < z < q + (y - q) = y.$$

If z were rational, $z = p$ with $p \in \mathbb{Q}$, then $r = k(p - q)$ would be rational, since κ preserves differences and products (the identities after Theorem 5.4.1, and its part (a)); this contradicts Example (The Class of the Newton Sequence). So z is irrational, and lies strictly between x and y.

The same argument works with r replaced by any positive irrational number. Between any two reals there are therefore both rationals and irrationals.

Exercise. C3: Density from the representatives

(a) Let $x = [a]$ and $y = [b]$ with $a, b \in \mathcal{C}$ and $x < y$. Prove, without Theorem 5.4.2 or Theorem 3.2.3, that there is $q \in \mathbb{Q}$ with $x < \kappa(q) < y$.

(b) Which hypothesis does your proof use that the proof of Theorem 5.4.3 in the text does not, and conversely?

Solution (Hint 1). Name the margin first: $b - a$ is positive. Then take the midpoint of a_N and b_N for one large N.

Solution (Hint 2). Fix δ and N_1 with $b_n - a_n \geq \delta$ for $n \geq N_1$, and thresholds N_2, N_3 for a and b at the tolerance $\delta/4$.

Solution.

(a) Write $x = [a]$ and $y = [b]$ with $a, b \in \mathcal{C}$. By Definition 5.3.2, $y - x = [b - a] \in P$, so $b - a$ is positive (Proposition 5.3.1). Fix $\delta \in \mathbb{Q}_{>0}$ and $N_1 \in \mathbb{N}$ with

$$b_n - a_n \geq \delta \quad \text{for every } n \geq N_1.$$

Then $\delta/4 = (\delta/2)/2 \in \mathbb{Q}_{>0}$ (Proposition 4.1.2 (b), twice). Since a and b are Cauchy in $\mathbb{Q}$, fix N_2 with $|a_m - a_n| < \delta/4$ for all $m, n \geq N_2$, and N_3 with $|b_m - b_n| < \delta/4$ for all $m, n \geq N_3$. Let

$$N := \max\{N_1, N_2, N_3\}, \qquad q := \frac{a_N + b_N}{2} \in \mathbb{Q}.$$

Let $n \geq N$. Expanding the right sides shows

$$q - a_n = (a_N - a_n) + \frac{b_N - a_N}{2},$$
$$b_n - q = (b_n - b_N) + \frac{b_N - a_N}{2}.$$

Since $n, N \geq N_2$, $a_N - a_n > -\delta/4$ (Proposition 4.1.3 (d)); since $N \geq N_1$, $b_N - a_N \geq \delta$, and multiplying by $1/2 > 0$ (Proposition 4.1.2 (b) with $\varepsilon = 1$) gives $(b_N - a_N)/2 \geq \delta/2$

(Proposition 2.4.3 (c), with equality allowed). Adding (Proposition 2.4.3 (b)),

$$q - a_n > -\frac{\delta}{4} + \frac{\delta}{2} = \frac{\delta}{4}.$$

In the same way, since $n, N \geq N_3$, $b_n - b_N > -\delta/4$, and $b_n - q > \delta/4$. So the sequences $(q - a_n)_n$ and $(b_n - q)_n$ are positive, each with margin $\delta/4$ and threshold N. By Definition 5.2.2 they represent $\kappa(q) - x$ and $y - \kappa(q)$. Hence $x < \kappa(q)$ and $\kappa(q) < y$. The rational q depends on the representatives a, b and on the constants fixed in the proof; nothing makes it the "best" rational, and a different run of the proof may give a different q.

(b) The proof in the text uses that $\mathbb{R}$ is Archimedean, through Theorem 4.2.5, and knows nothing of representatives. Your proof in (a) uses the Cauchy property of a and b and positivity of $b - a$, and no Archimedean property of $\mathbb{Q}$ or $\mathbb{R}$.

Marking scheme. (10 points) - 2: the margin δ from positivity of $b - a$, named before any index. - 3: Cauchy thresholds at a tolerance small enough for the estimates (relative to δ), and one N beyond every threshold used. - 3: both estimates $q - a_n > \delta/4$ and $b_n - q > \delta/4$ for every $n \geq N$. - 2: the estimates read as positivity of two representatives, of $\kappa(q) - x$ and $y - \kappa(q)$. Fatal errors (0 for the step): a midpoint of a_n and b_n with n varying; q compared only with a_N and b_N, not with every later term.